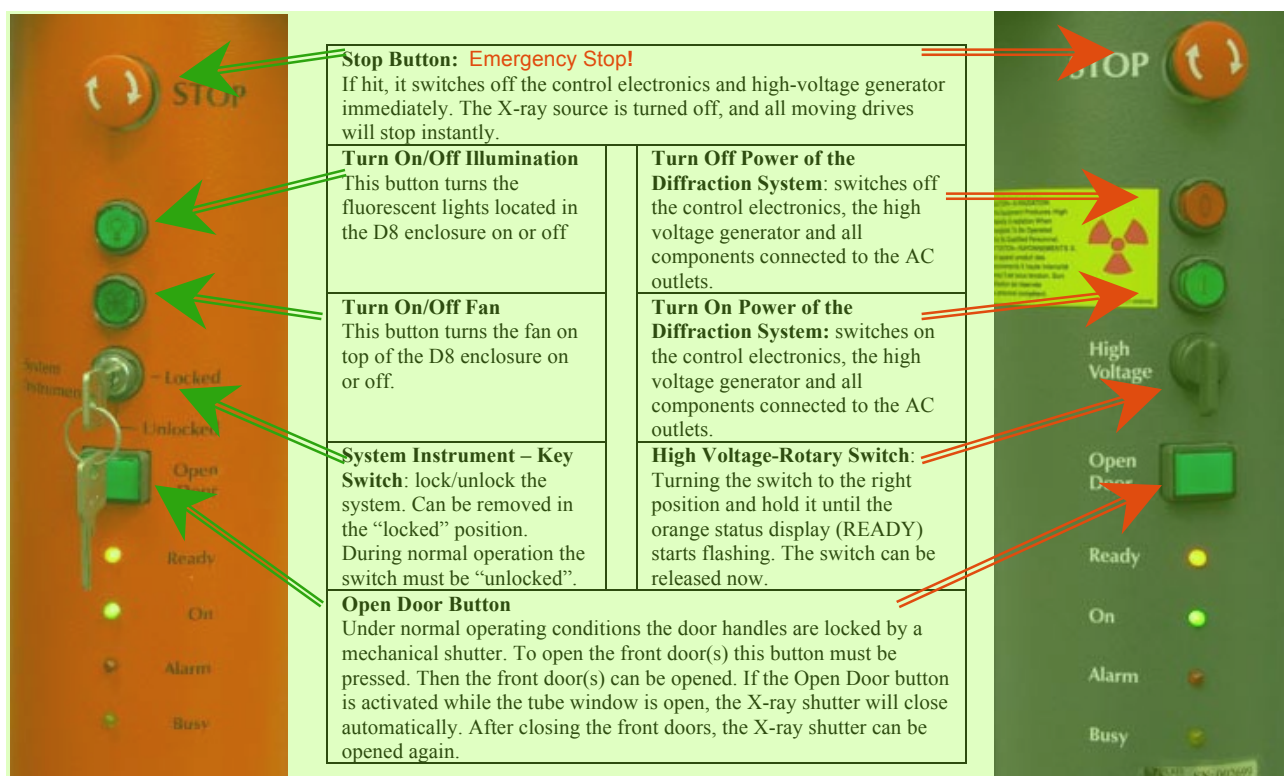


Bruker D8 Discover XRD User's Guide 1

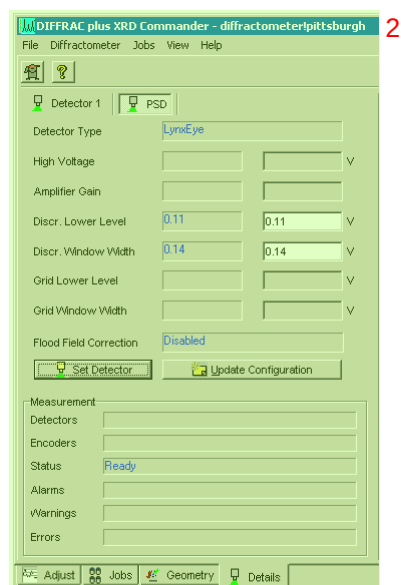
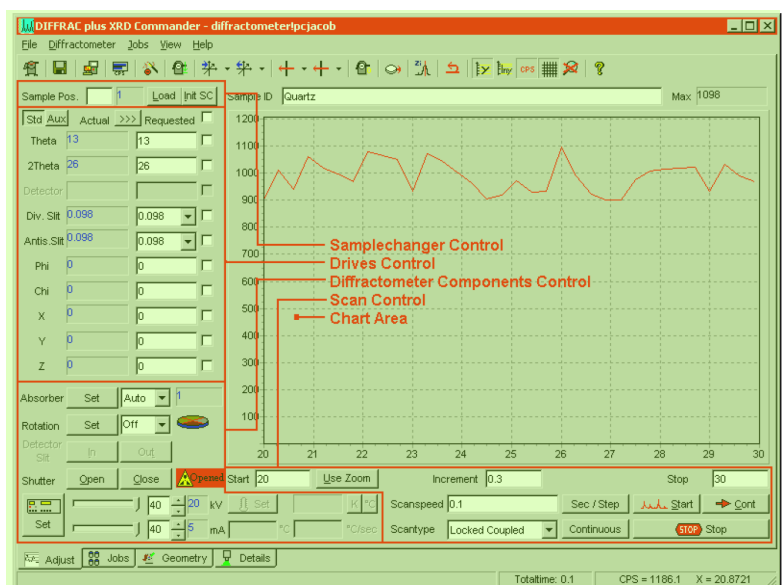
I. Powder Diffraction Measurement using LynxEye Detector

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1. Fill in the Log-book with relevant information before you begin analyze your sample. 3
2. Check System Control Buttons (located at the front side of the system) and make sure the Ready and On LED lights are on. 4



3. Before opening the D8 door, check for any abnormalities in and around the instrument. Give particular attention to the **shutter lights on the X-ray tube**. The **green light** indicates that the tube is energized (light should be lit). The two **red lights** indicate the **shutter position** (*if they are lit the shutter is open and you must not open the doors on the D8; if the red lights are off, then it is safe to open the doors and mount the sample stage on the goniometer*). 6
4. Mount sample in the appropriate holder and then place the holder in the sample stage on the goniometer. 7
5. If the **XRD COMMANDER** is not opened, click the icon,  on the desktop to open it. 8
6. Set up DIFFRACplus XRD Commander 9
- 10



1) Click the **Details** tab to activate the **Details** page, 3

a. Select the **PSD detector (LynxEye)**, 4

b. Input **Discriminator Lower Level: 0.12; Discriminator Window Width: 0.14**, 5

c. Click **Set Detector** button, 6

2) Activate the **Adjust** page 7

a. Make sure that for each drive its actual position is shown in **blue**. If any of the drive positions are displayed in dark red, you will not be able to use the D8 — **ask for help**. 8

b. Set the X-ray generator voltage to **40 kV** and the current to **40mA** or other your desired combination. 9

c. Find zero position of the **Z** drive for your sample: 10

i. Open the Video program by clicking the video icon,  11, and start live view. 12

ii. In **XRD Commander** window, click **Direct Mode** icon,  to enter direct laser control mode. 13

iii. In **Direct Command** window, type **RC1** and press **Enter** to activate the direct laser control mode. 14

iv. Type **OC16,1** to turn the laser on. 15

v. Close the **Direct Command** window. 16

vi. Input a specific **Z** position and click the **drive to** icon,  to adjust the **Z** drive to center the laser spot at the crosshair. 17

vii. Open the **Direct Command** window again, and type **OC16,0** to turn off the laser. 18

viii. In Video program, stop Grab the live video. 19

3) Select measurement parameters: 20

a. scan type: **Locked Coupled, continuous** scan. 21

b. start: **>3.5°**, stop: **<100°**, scanspeed: **0.40** Sec/Step, and increment: **0.04°**. 22

c. slit size: **0.2** mm; high voltage: **40** kV, current: **40** mA. 23

4) Click **Start** button,  to begin a measurement. 24

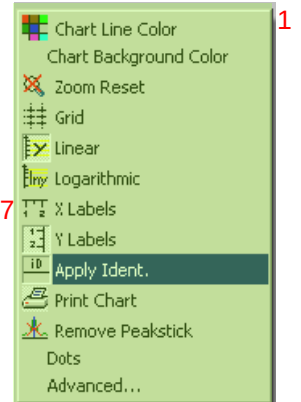
5) Click **Stop** button,  to end the immediate measurement. 25

7. After the scan is complete 26

1) Save the measured data as a **RAW** data file to your folder on the hard drive. **XRD Commander** displays the **Save As** dialog box, and you must enter the **RAW** filename. The **RAW** data file can later be examined (e. g., with EVA). 27

Note: by clicking the *right mouse button* inside the **Chart** area, you will bring a 2 popup menu:

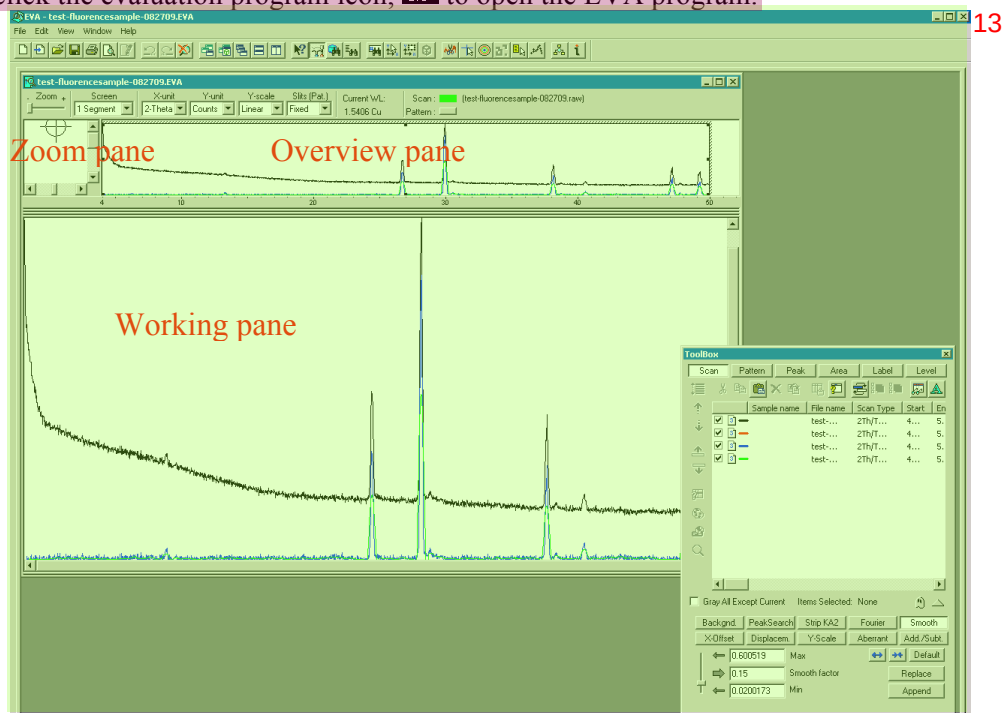
- **Apply Ident** (applies the sample identification on the top axis) 3
- **Print Chart** 4
- **Remove Peakstick** 5
- **Dots** (shows the scan data as little dots instead of connected lines) 6
- **Advanced** (shows advanced customized options for the chart area such as 7
Save the chart data as a Text, XML, HTML, or Excel file).
Export→Data→Excel→Save As to your folder on the hard drive. 8



- 2) Set the X-ray tube voltage to 20 kV, current to 5mA 9
- 3) **Minimize XRD Commander. Do not close the program.** 10

II. Diffraction Evaluation using *DIFFRAC^{plus} BASIC* Evaluation 11

1. Double click the evaluation program icon,  to open the EVA program. 12

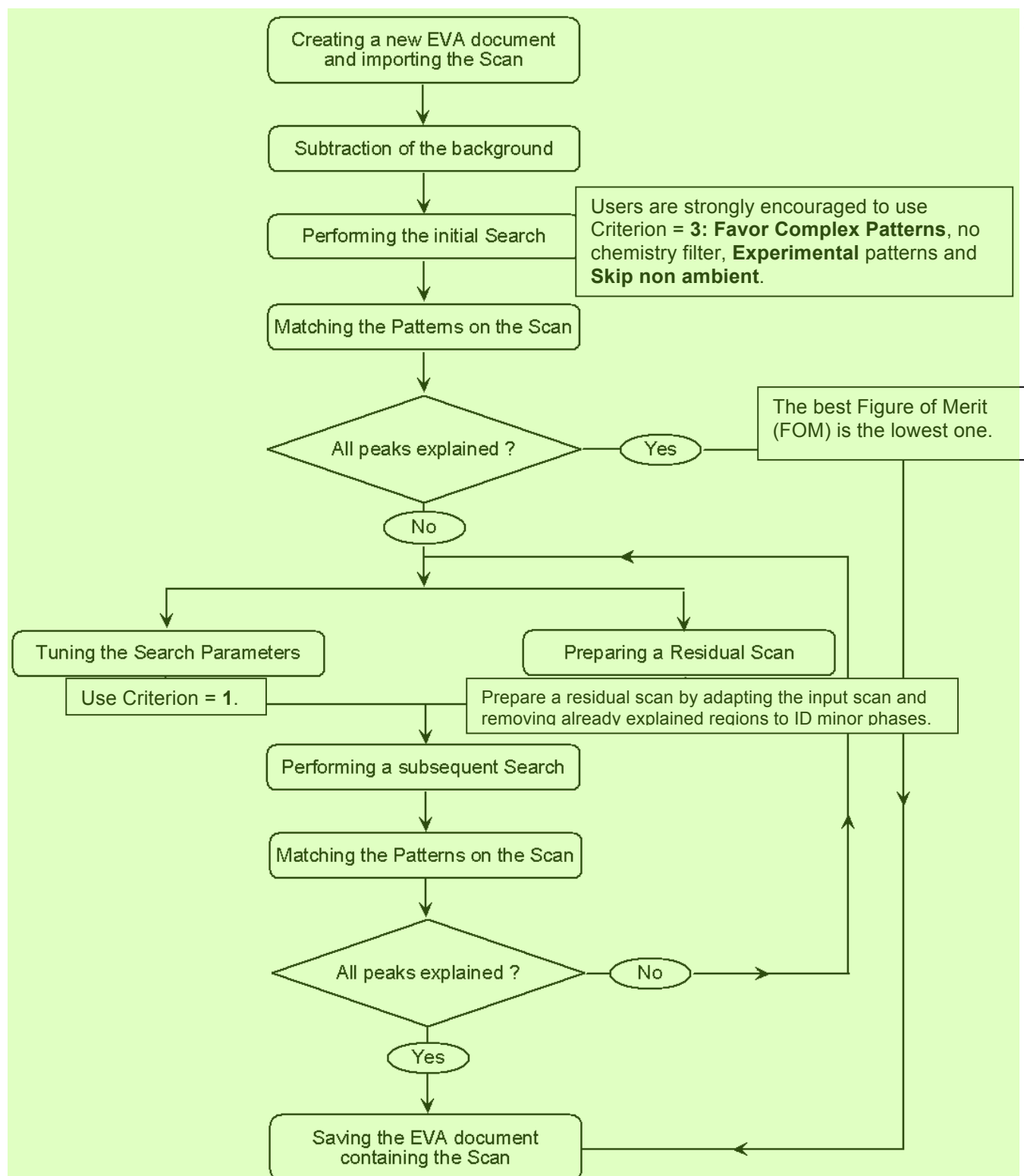


2. Creating EVA Documents 14
 - a. On the toolbar, click the **Import** button. 15
 - b. Locate the data **.RAW** file you want to import in the directory of your RAW files. 16
 - c. Click **Open**. 17
3. Working with Scans 18
 - a. Performing a Background Subtraction 19
 - i. The **DIFFRAC** traditional method: *prepare data for search/match*. 20
 - 1) Click the **Background** button on the Data Treatment palette 21
 - 2) Use the slider to adjust the curvature if there are background humps. 22
 - 3) Select the **Subtract** from Scan check box. 23
 - 4) Click **Append** to append the background-subtracted scan data to the document 24
 - ii. An enhanced method: gives a smooth curve.
 - iii. A **Bézier** method: allows you to draw the background exactly the way you want. 25
 - 1) Draw a calculated background with the method above; get as close as you can 26 to the background you intend to draw.
 - 2) Check the **Bézier** box to transform this background into curves. 27

- 3) Move the *slider* to adjust the curve to what you intend to draw. 1
- 4) Click **Edit** button to display all the control points: the passing points are disks, 2 the tangent points are squares, and the tangent vectors are in dashed lines. You can **move**, **erase**, or **add** a passing point.
- 5) Check the **Subtract** box and then click on **Append** to create your drawn 3 background.
- b. Performing a Peak Search 4
 - i. On the Data Treatment palette click the Peak Search button. 5
 - ii. Move the slider to see ghost peaks in the Overview and Working panes. 6
 - iii. When no manual editing is required, click **Make DIF** to transfer the found peaks to a 7 DIF pattern.
 - iv. When manual editing is required, click **Append To List**. 8
- c. Computing $K_{\alpha 2}$ Stripping 9
 - i. Click the **Strip $K_{\alpha 2}$** button on the Data Treatment palette. 10
 - ii. Use the slider to adjust the Intensity Ratio. 11
 - iii. Click Append to append the $K_{\alpha 2}$ subtracted scan to the document 12
- d. Smoothing Scans 13
 - i. Click the Smooth button on the Data Treatment palette. 14
 - ii. Use the slider to adjust the Smooth factor. 15
 - iii. Click Append to append the smoothed scan to the document 16
- e. Fourier Smoothing and Expansion 17
 - i. Click the Fourier button on the Data Treatment palette. 18
 - ii. In the Fourier treatment control panel, select the expansion (x1, x2, x4, x8, and x16). 19
 - iii. Use the slider to adjust the cutoff. 20
 - iv. Click Append to append the smoothed scan to the document. 21
4. Search/Match:  22 (** see the next page for a general procedure.) 23
5. Saving and Printing in EVA. 24
6. Export file as ASCII file. 25
 - a. **Export** your EVA file as a RAW file; 26
 - b. Open the **RAW File Exchange**  28, or **XCH**  29, to convert your RAW and/or processed 27 files to ASCII files.
7. Exit EVA after you complete data processing. 30
8. Sign off the instrument by filling in the log-book. 31

** The purpose of EVA **Search/Match** is to search the current scan of an unknown material and then identify reference patterns that are likely to explain the unknown scan.

The general procedure for EVA Search/Match is as follows:



Use the **Scroll List** mode to match the scan 3